Namal University, Mianwali

Department of Computer Science

System Design of Project

Milestone 3

Course Code	CSC-225
Course Name	Software Engineering
Instructor	MS. Asiya Batool

Group Members		
Name	**Roll No**	**Email**
Abdul Rehman	01	bscs24f01@namal.edu.pk
Junaid Gondal	28	bscs24f28@namal.edu.pk
Ahmad Ali	02	bscs24f02@namal.edu.pk

Submission Date: January 18, 2026

Contents

1 Introduction **2**

2 Design Assumptions and Constraints **2**

2.1 Design Assumptions 2

2.2 Design Constraints 2

3 Key Design Decisions **3**

4 System Design Diagrams **3**

4.1 Use Case Diagram 3

4.2 DFD Level 0 4

4.3 DFD Level 1 5

4.4 Activity Diagrams 6

4.4.1 Student Activity Diagram 6

4.4.2 Teacher Activity Diagram 7

4.4.3 Administrator Activity Diagram 8

4.5 Sequence Diagrams 9

4.6 Class Diagram 12

4.7 Component Diagram 13

5 Requirements–Design Traceability Table **14**

6 Conclusion **18**

1 Introduction

As part of Project Milestone 3, the **System Design Report** for the Fair and Unbiased Assessment System (FUAS) is presented in this document. This milestone aims to create a comprehensive and organized system design based on the approved requirements specified in the Software Requirements Specification (SRS).

The report describes how various users operate the system, how data moves through the system, and how the system's components interact. Every design component follows carefully to the scope specified in the SRS and does not add any extra or irrelevant features.

2 Design Assumptions and Constraints

2.1 Design Assumptions

Realistic assumptions derived from the SRS form the basis of the system design. Users are presumed to be familiar with websites. In order to facilitate automated evaluation, student submissions must be reasonably readable. During assessment checking, it is assumed that the AI evaluation engine is accessible.

- Users possess basic computer literacy
- Handwritten submissions are readable in **PDF files**
- AI evaluation engine remains operational
- Internet connectivity is available
- Teachers upload correct answer keys

2.2 Design Constraints

Performance constraints, security requirements, and institutional policies all limit the design. The system must support multiple users during periods of high usage while guarantees safe data handling and restricted access.

- Compliance with institutional data protection policies
- Role Based Access Control
- Secure storage of credentials and assessment data
- Support for concurrent users

3 Key Design Decisions

Several design decisions were made to ensure system reliability and fairness. A layered architecture was selected to separate concerns and improve maintainability. A centralized evaluation engine ensures unbiased grading, while role specific dashboards enhance usability.

- Layered system architecture
- Centralized automated evaluation engine
- Role-based dashboards
- Encrypted authentication mechanisms

4 System Design Diagrams

4.1 Use Case Diagram

The use case diagram illustrates the core functionalities of the Fair & Unbiased Assessment System (FUAS) for three main actors:

Key Points

- Students can upload answers, view reports, and see class performance.
- Teachers create assessments, upload answer keys, generate reports, and control marks visibility.
- Administrators manage user credentials, enroll students, and assign courses.
- System relationships show dependencies (include) and optional features (extend).

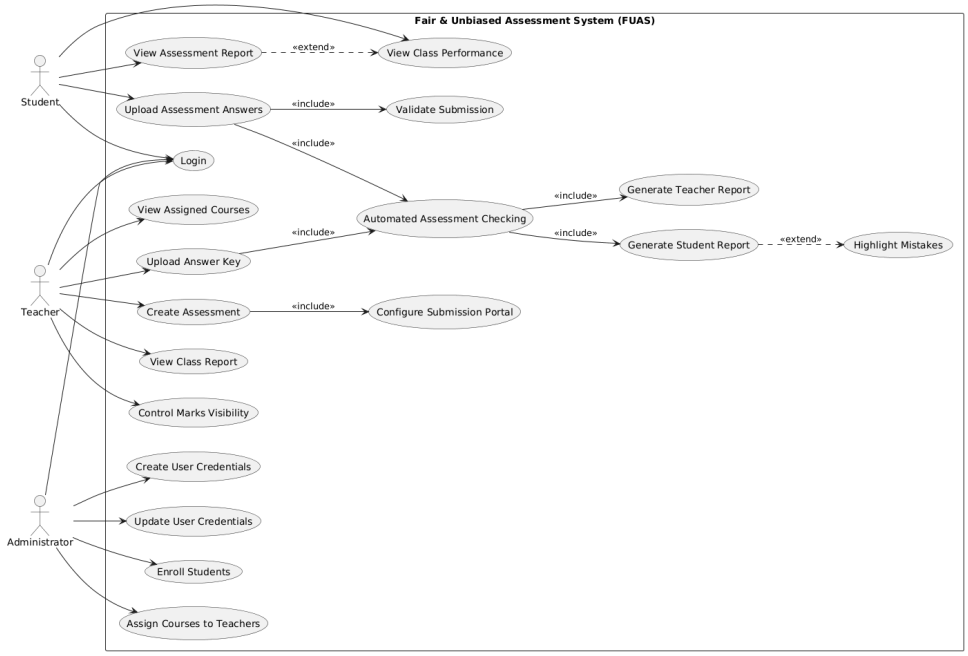

Figure 1: DFD Level 0 of Fair and Unbiased Assessment System

4.2 DFD Level 0

The Level 0 Data Flow Diagram presents the system as a single process interacting with external entities. It highlights the system boundary and major data exchanges without internal details.

- External Entities: Student, Teacher, Administrator
- Central Process: FUAS
- Data Flows: Credentials, submissions, reports etc.

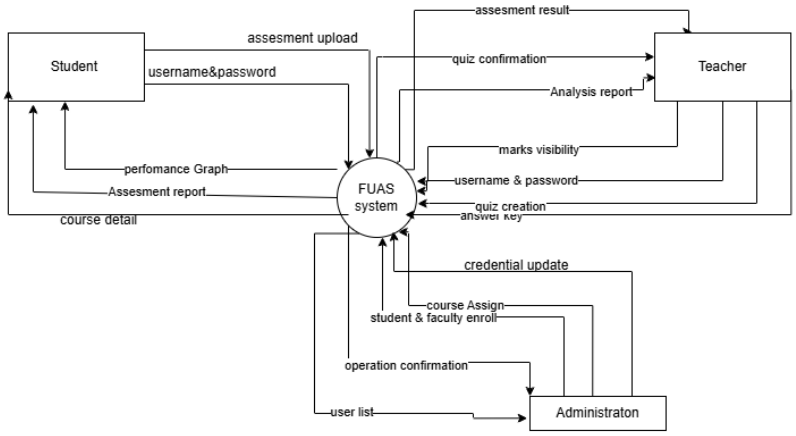

Figure 2: DFD Level 0 of Fair and Unbiased Assessment System

4.3 DFD Level 1

The Level 1 DFD do deep into the system into internal processes and shows how data flows between modules and data stores.

- User Authentication
- Course Management
- Assessment portal creation Submission
- Automated Evaluation
- Answer Submission handling
- Report Generation

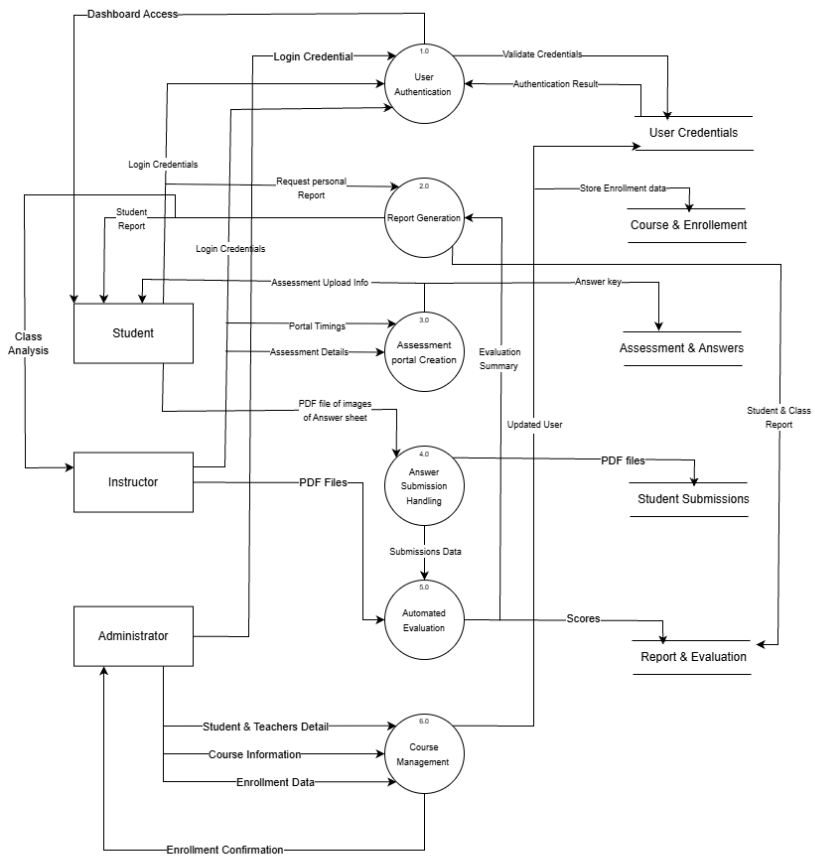

Figure 3: DFD Level 1 of Fair and Unbiased Assessment System

4.4 Activity Diagrams

This section describes the activity diagrams of the Fair and Unbiased Assessment System (FUAS). Activity diagrams represent the dynamic behavior of the system by illustrating the sequence of actions performed by different system actors. The activities are categorized according to the three primary user roles: Student, Teacher, and Administrator.

4.4.1 Student Activity Diagram

The Student Activity Diagram illustrates the workflow followed by a student while interacting with the system from login to logout.

Key Activities

- Login to the system using institutional credentials.
- Access the student dashboard after successful authentication.
- Check the availability of quizzes, assignments, or assessments.
- Upload handwritten answer sheets in image or PDF format.
- Wait for the automated assessment evaluation process.
- View detailed assessment reports generated by the system.
- Analyze class performance graphs if marks visibility is enabled.
- Logout from the system after completing activities.

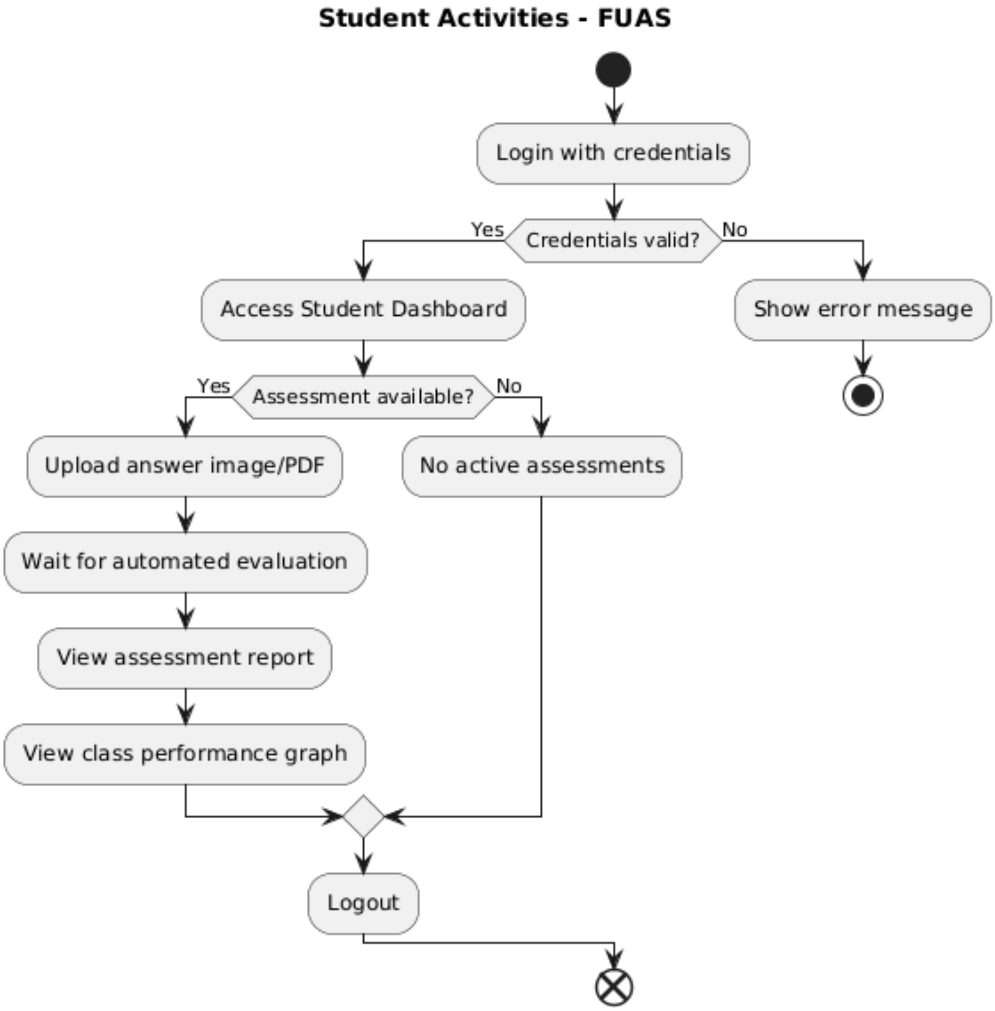

Figure 4: Student Activity Diagram of the Fair and Unbiased Assessment System

4.4.2 Teacher Activity Diagram

The Teacher Activity Diagram represents the sequence of tasks performed by a teacher to create, manage, and evaluate assessments.

Key Activities

- Login to the system using teacher credentials.
- Access the teacher dashboard to manage assigned courses.
- Select a course and create a new assessment.
- Define submission portal start and end timings.
- Upload answer keys for automated evaluation.
- Enable or disable student submission options.
- Review system-generated class performance reports.

- Control the visibility of marks for students.
- Logout from the system after task completion.

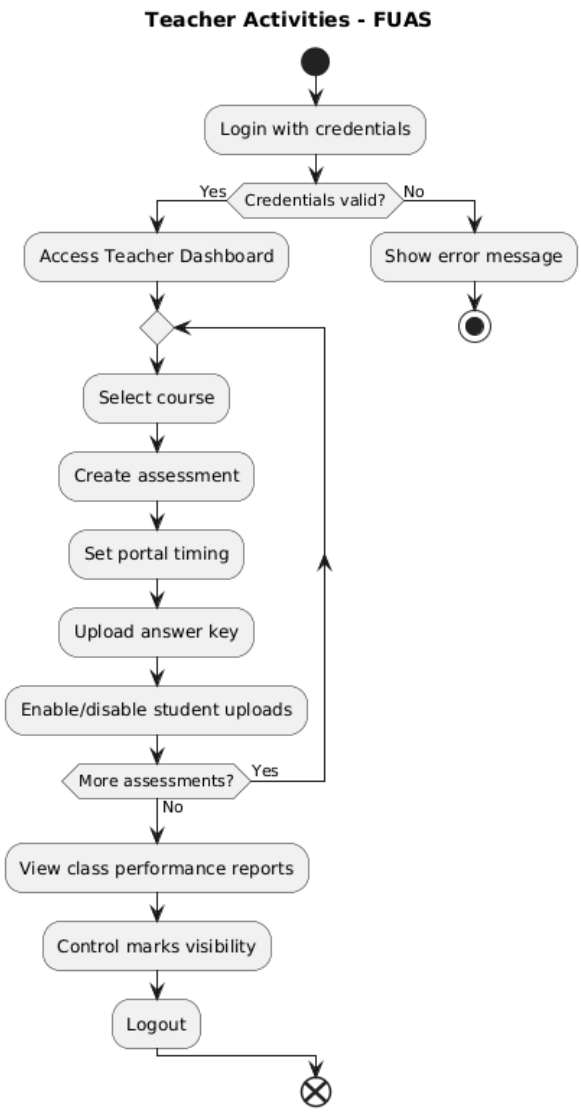

Figure 5: Teacher Activity Diagram of the Fair and Unbiased Assessment System

4.4.3 Administrator Activity Diagram

The Administrator Activity Diagram describes the administrative workflows required for system configuration, user management, and monitoring.

Key Activities

- Login to the system using administrator credentials.
- Access the administrator dashboard.
- Assign courses to faculty members.

- Create and manage faculty login credentials.
- Enroll students into courses.
- Generate and update student credentials.
- Manage system configurations and settings.
- Monitor system logs and performance metrics.
- Logout from the system securely.

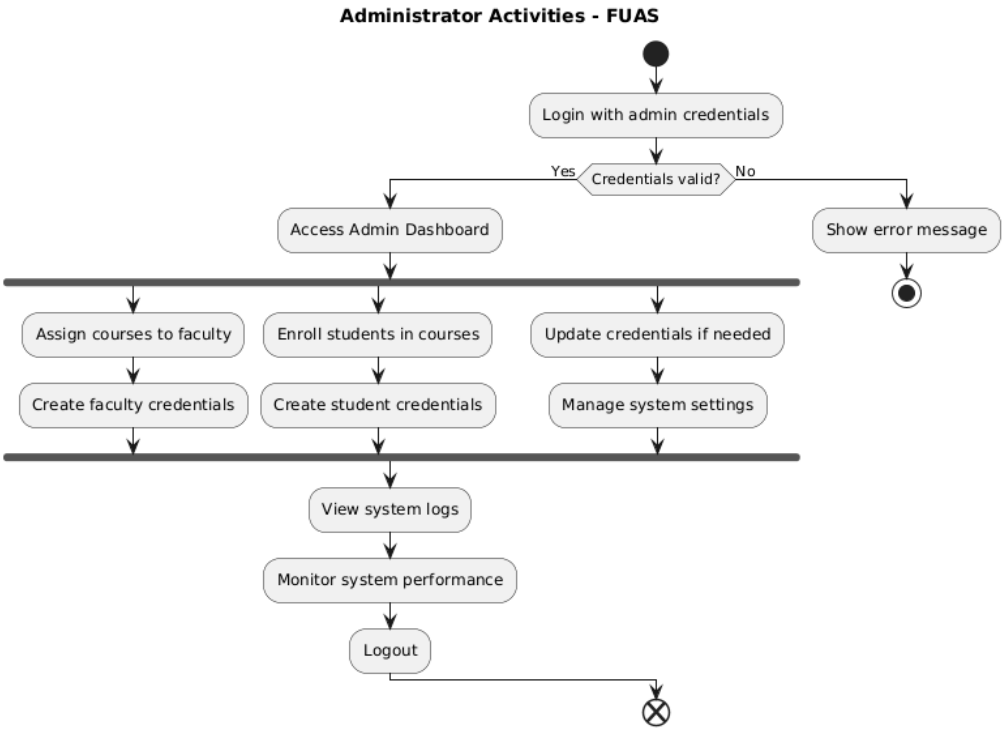

Figure 6: Administrator Activity Diagram of the Fair and Unbiased Assessment System

4.5 Sequence Diagrams

Sequence diagrams describe the interaction between users and system components over time.The Student Sequence Diagram shows authentication, assessment upload, storage, automated evaluation, and report viewing. The Teacher Sequence Diagram represents assessment creation, answer key upload, triggering evaluation, and viewing class reports. The Administrator Sequence Diagram demonstrates credential management and course assignment processes.

These diagrams ensure clarity in message flow and system response for each user role.

- Student sequence
- Teacher sequence

- Administrator sequence

Figure 7: Student Sequence Diagram

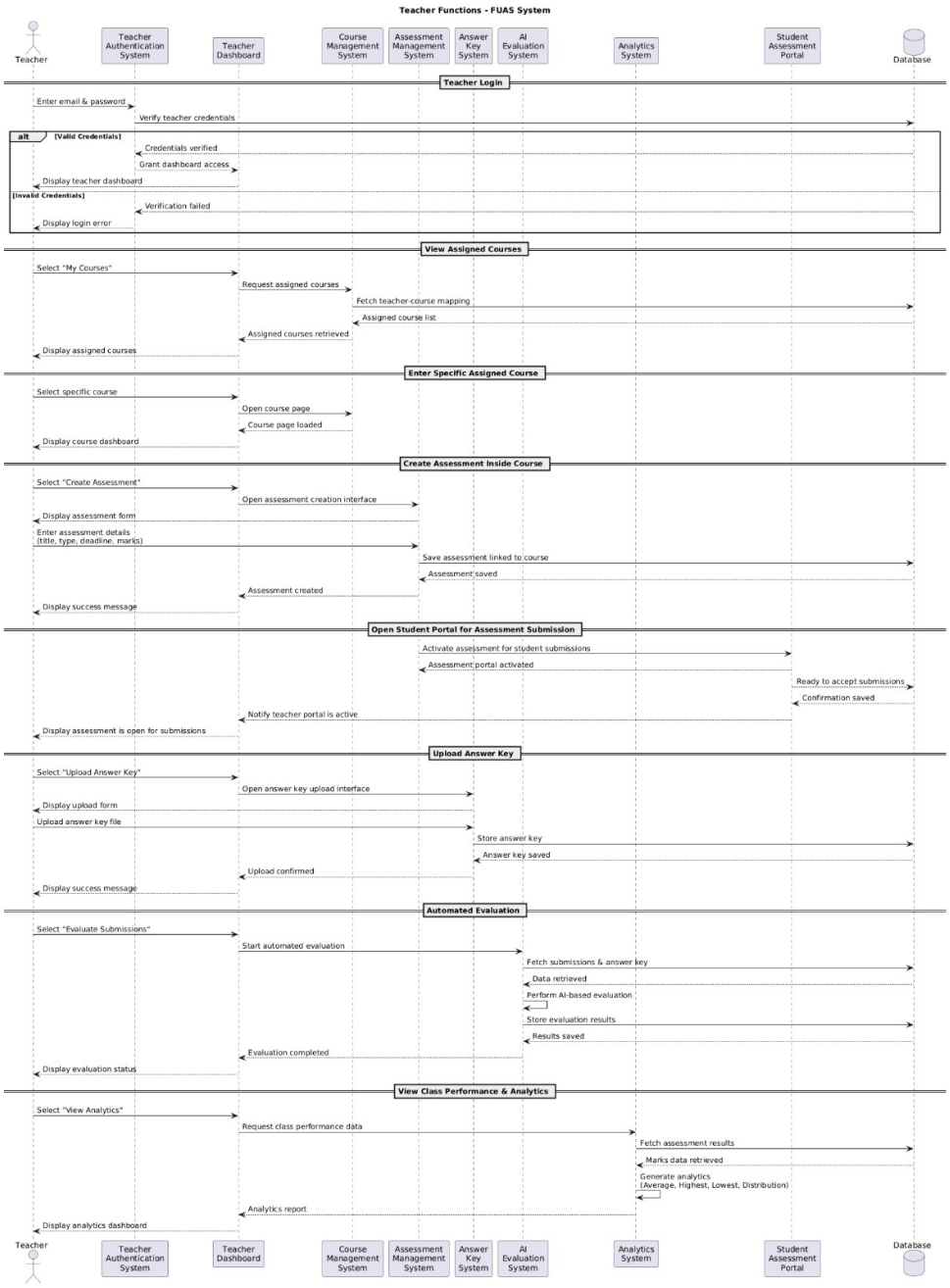

Figure 8: Teacher Sequence Diagram

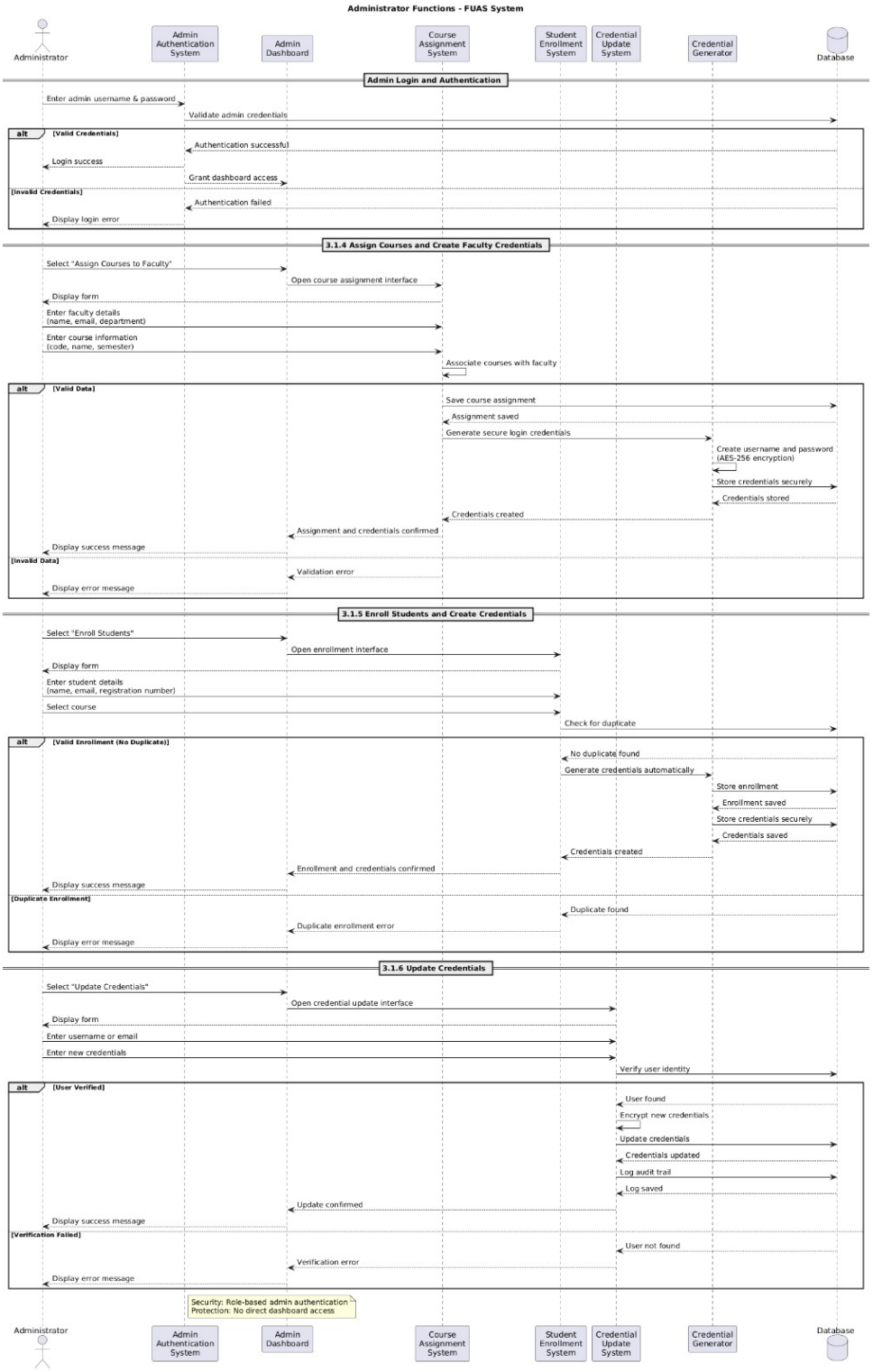

Figure 9: Administrator Sequence Diagram

4.6 Class Diagram

The class diagram represents the static structure of the system and relationships among system entities.

- User, Student, Teacher, Administrator

- Course, Assessment, Submission
- Report, EvaluationEngine

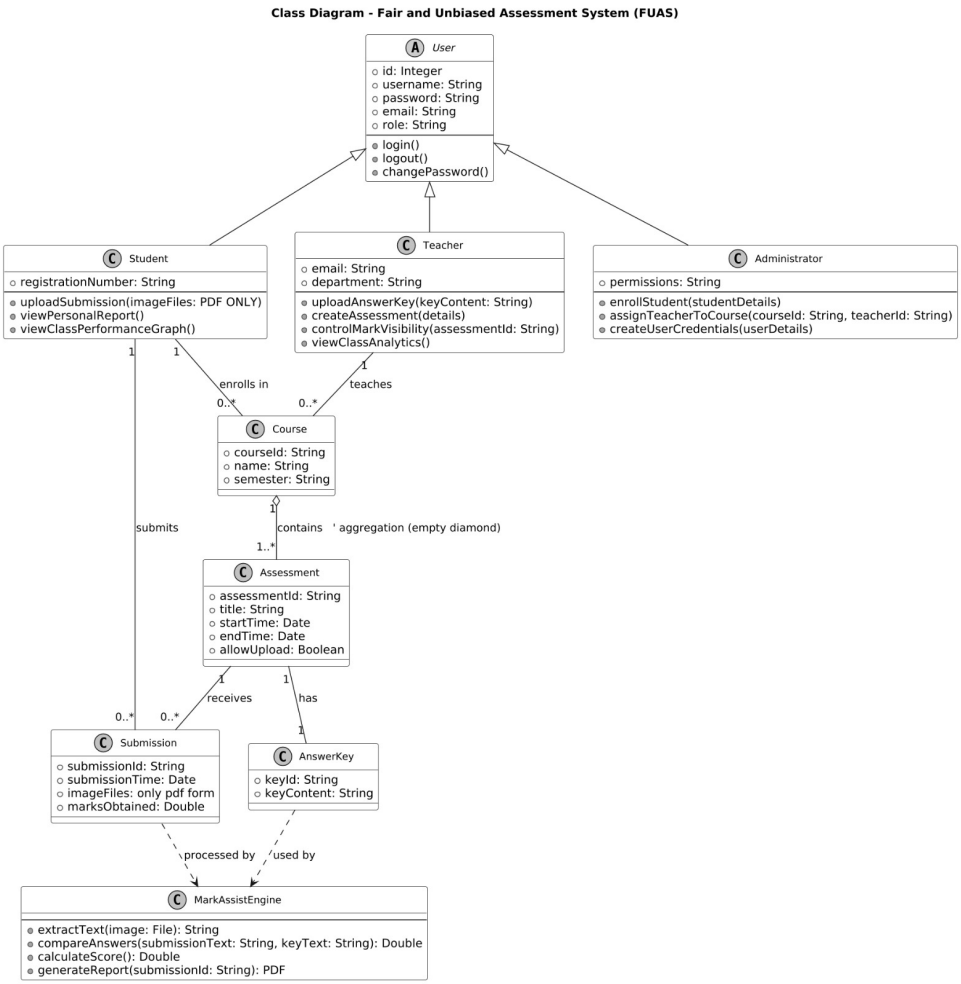

Figure 10: Class Diagram of FUAS

4.7 Component Diagram

The component diagram presents the high-level architectural organization of the system.

- User Interface
- Authentication Module
- Assessment Management Module
- Evaluation Engine
- Reporting Module
- Database

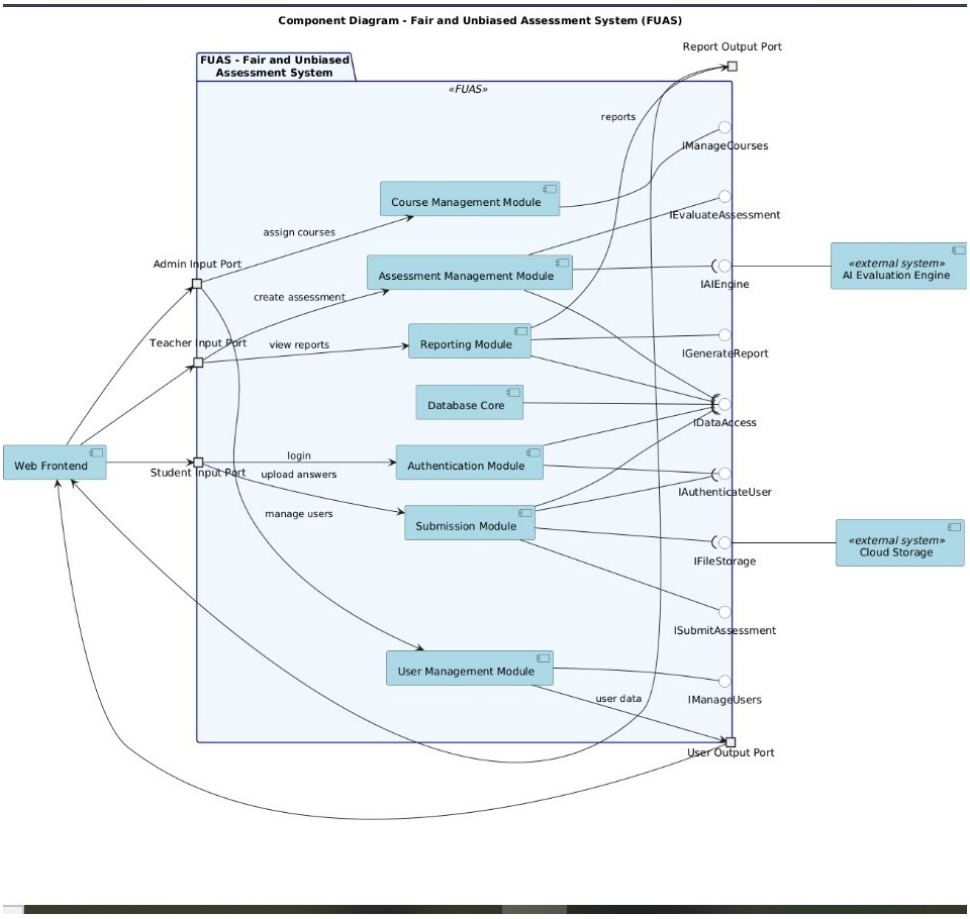

Figure 11: Component Diagram of FUAS

5 Requirements–Design Traceability Table

Requirement ID	Description	Design Artifact
FR-1	User Authentication (Student, Teacher, Administrator)	Use Case Diagram, DFD Level 0, DFD Level 1, Component Diagram, Student Sequence Diagram, Teacher Sequence Diagram, Administrator Sequence Diagram, Student Activity Diagram, Teacher Activity Diagram, Administrator Activity Diagram

Requirement ID	Description	Design Artifact
FR-2	Student Login with Registration Number and Password	Use Case Diagram, Student Sequence Diagram, Student Activity Diagram, Class Diagram, Component Diagram
FR-3	Teacher Login with Email and Password	Use Case Diagram, Teacher Sequence Diagram, Teacher Activity Diagram, Class Diagram, Component Diagram
FR-4	Administrator Login with Admin Credentials	Use Case Diagram, Administrator Sequence Diagram, Administrator Activity Diagram, Class Diagram, Component Diagram
FR-5	Create User Credentials (Administrator)	Use Case Diagram, Administrator Sequence Diagram, Administrator Activity Diagram, DFD Level 1, Component Diagram
FR-6	Update User Credentials (Administrator)	Use Case Diagram, Administrator Sequence Diagram, Administrator Activity Diagram, DFD Level 1, Component Diagram
FR-7	Enroll Students in Courses (Administrator)	Use Case Diagram, Administrator Sequence Diagram, Administrator Activity Diagram, DFD Level 1, Component Diagram
FR-8	Assign Courses to Teachers (Administrator)	Use Case Diagram, Administrator Sequence Diagram, Administrator Activity Diagram, DFD Level 1, Component Diagram, Class Diagram

Requirement ID	Description	Design Artifact
FR-9	View Assigned Courses (Student & Teacher)	Use Case Diagram, Student Activity Diagram, Teacher Activity Diagram, Class Diagram, DFD Level 1
FR-10	Create Assessment (Teacher)	Use Case Diagram, Teacher Sequence Diagram, Teacher Activity Diagram, DFD Level 1, Component Diagram, Class Diagram
FR-11	Configure Submission Portal Timing (Teacher)	Use Case Diagram, Teacher Activity Diagram, Teacher Sequence Diagram, Component Diagram, Class Diagram
FR-12	Upload Answer Key (Teacher)	Use Case Diagram, Teacher Sequence Diagram, Teacher Activity Diagram, DFD Level 1, Component Diagram, Class Diagram
FR-13	Enable/Disable Student Uploads (Teacher)	Use Case Diagram, Teacher Activity Diagram, Class Diagram
FR-14	Upload Assessment Answers (Student - PDF/Image)	Use Case Diagram, Student Sequence Diagram, Student Activity Diagram, DFD Level 0, DFD Level 1, Component Diagram, Class Diagram
FR-15	Validate Submission (System)	Use Case Diagram, Student Sequence Diagram, Student Activity Diagram, DFD Level 1, Component Diagram
FR-16	Store Student Submissions Securely	DFD Level 1, Student Sequence Diagram, Component Diagram, Class Diagram

Requirement ID	Description	Design Artifact
FR-17	Automated Assessment Checking (AI Engine)	Use Case Diagram, DFD Level 1, Student Sequence Diagram, Teacher Sequence Diagram, Component Diagram, Class Diagram
FR-18	Compare Student Answers with Answer Key	Component Diagram, Class Diagram, DFD Level 1
FR-19	Calculate Scores and Marks	DFD Level 1, Component Diagram, Class Diagram, Student Sequence Diagram
FR-20	Generate Student Report	Use Case Diagram, Student Sequence Diagram, Student Activity Diagram, DFD Level 1, Component Diagram, Class Diagram
FR-21	Generate Teacher Report	Use Case Diagram, Teacher Sequence Diagram, Teacher Activity Diagram, DFD Level 1, Component Diagram
FR-22	View Assessment Report (Student)	Use Case Diagram, Student Sequence Diagram, Student Activity Diagram, DFD Level 0, Class Diagram
FR-23	View Class Performance (Student)	Use Case Diagram, Student Sequence Diagram, Student Activity Diagram, Class Diagram
FR-24	View Class Report (Teacher)	Use Case Diagram, Teacher Sequence Diagram, Teacher Activity Diagram, DFD Level 1, Component Diagram
FR-25	Highlight Mistakes in Student Submission	Use Case Diagram, Class Diagram, Component Diagram

Requirement ID	Description	Design Artifact
FR-26	Control Marks Visibility (Teacher)	Use Case Diagram, Teacher Activity Diagram, Teacher Sequence Diagram, Class Diagram
FR-27	Secure Data Storage (Credentials, Submissions, Reports)	DFD Level 1, Component Diagram, Class Diagram
FR-28	Role-Based Access Control	Component Diagram, Class Diagram, All Sequence Diagrams
FR-29	Logout Functionality (All Users)	Use Case Diagram, Student Activity Diagram, Teacher Activity Diagram, Administrator Activity Diagram
FR-30	Dashboard Access (Student, Teacher, Administrator)	DFD Level 0, Student Sequence Diagram, Teacher Sequence Diagram, Administrator Sequence Diagram, All Activity Diagrams

6 Conclusion

A thorough system design for the Fair and Unbiased Assessment System is presented in this report. Every design component is taken straight from the SRS and stays inside the project's specified parameters. Fairness, security, scalability, and maintainability are all guaranteed by the design, which offers a solid basis for implementation in upcoming milestones.